SCALABILITY & FUTURE PLAN

Date: 20 October24
Team ID: Roll 58
Project Name: EV Market Analysis Dashboard using Tableau Public
Maximum Marks: 1 Mark

Scalability & Future Plan

This document describes how the EV Market Analysis Dashboard can be expanded to handle larger datasets, more users, and additional analytical features. The current project uses the ev_market_train.csv dataset, containing 200,000 records and 37 columns in the previously inspected version. Future enhancements will focus on improving dashboard performance, extending data storage capabilities, protecting data, and providing more advanced EV market insights.

Current System Limitations

S.No.	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The dashboard depends on the currently available EV dataset.	New vehicles, updated sales figures, and changing market trends may not be reflected automatically.	High
2	A large dataset with multiple visualizations and filters may increase loading or response time.	Users may experience slower dashboard interactions as data volume and complexity grow.	High
3	Tableau Public is intended for public sharing, and published workbooks may be publicly accessible.	Sensitive or restricted information should not be included in a public workbook.	Medium

Scalability Plan

S.No.	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	The dashboard is designed for users exploring EV market data.	Improve dashboard organization and consider an access-controlled analytics platform if private, concurrent multi-user access becomes necessary.
2	Data Storage	The project uses a CSV dataset containing EV records.	Integrate a database or cloud data warehouse when the dataset expands or requires regular updates.
3	Performance	Dashboard performance depends on dataset size,	Optimize calculations, reduce unnecessary fields, use efficient data extracts or aggregations, and test filter response times.

		calculations, and visualization complexity.	
4	Security	Tableau Public supports public dashboard sharing.	Review the data before publishing and use a platform with access controls if confidential data or restricted access is required.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve dashboard layout, filters, tooltips, and visualization readability.	Within 1 month	Easier navigation and improved user experience.
Phase 2	Integrate updated EV datasets and automate data refresh where supported.	Within 2–3 months	More current and comprehensive EV market analysis.
Phase 3	Connect a database or cloud data warehouse to manage larger datasets.	Within 3–6 months	Better support for growing data volumes and repeatable analysis.
Phase 4	Add advanced analytics, such as EV sales forecasting and vehicle segmentation.	Within 6–9 months	Deeper insights into possible market trends and vehicle groups.
Phase 5	Introduce controlled access and enhanced security if the project expands to restricted or organizational use.	As required	Better access management and protection of non-public information.

Expected Future Outcome

The proposed scalability plan will help evolve the EV Market Analysis Dashboard from a descriptive visualization project into a more comprehensive EV analytics solution. By introducing updated data sources, performance improvements, advanced analytical features, and suitable access controls, the system can support broader analysis as project requirements grow.

Note: The timelines are proposed estimates, not committed delivery dates. Future features have not been claimed as already implemented.